

```
CREATE DATABASE CompanyDB;  
USE CompanyDB;
```

```
CREATE TABLE employees (  
    id INT,  
    name VARCHAR(50),  
    department VARCHAR(50),  
    salary INT,  
    age INT,  
    city VARCHAR(50)  
);
```

```
INSERT INTO employees VALUES  
(1,'John','IT',60000,28,'London'),  
(2,'Sara','HR',45000,32,'New York'),  
(3,'Mike','IT',70000,30,'London'),  
(4,'Anna','Finance',65000,27,'Toronto'),  
(5,'David','HR',40000,35,'Sydney'),  
(6,'Emma','IT',72000,29,'New York');
```

Q1. Retrieve all employees from the table.

```
//SELECT * FROM employees;
```

Q2. Display only employee names and their departments.

```
//SELECT name,department FROM employees;
```

Q3. Find employees who work in the IT department.

```
//SELECT * FROM employees WHERE department='IT';
```

Q4. Show employees who earn more than 60000.

```
//SELECT * FROM employees WHERE salary>60000;
```

Q5. Display employees who live in London.

```
// SELECT * FROM employees WHERE city = 'London';
```

Q6. Sort employees by salary in ascending order.

```
//SELECT * FROM employees ORDER BY salary ASC;
```

Q7. Sort employees by salary in descending order.

```
//SELECT * FROM employees ORDER BY salary DESC;
```

Q8. Display the first 3 employees from the table.

```
//SELECT * FROM employees LIMIT 3;
```

Q9. Find employees whose age is between 28 and 32.

```
//SELECT * FROM employees WHERE age BETWEEN 28 AND 32;
```

```
//SELECT * FROM employees WHERE age >= 28 AND age <= 32;
```

Q10. Find employees who work in either IT or HR departments.

```
//SELECT * FROM employees WHERE department IN('IT','HR');
```

```
//SELECT * FROM employees WHERE department = 'IT' OR department = 'HR';
```

Q11. Find employees whose name starts with 'A'.

```
//SELECT * FROM employees WHERE name LIKE 'A%';
```

[illegible]

SELECT Summary

////////////////////////////////////

DQL - Data Query Language

Function	SQL Keyword/Syntax
Select all columns	SELECT *
Select specific columns	SELECT <column>
Filter rows	WHERE
Sort results	ORDER BY
Limit the number of rows returned	LIMIT
Specify a range of values	BETWEEN
Match multiple possible values	IN
Search for a pattern	LIKE

////////////////////////////////////

Activity 1 - Student Information System

A small institute wants to store basic information about students.

Each student has an **ID, name, course, age, and city.**

Create a database and a table to store this information and insert some sample data.

Tasks

1. Create a database named **StudentDB**.
2. Create a table called **students** with the following fields:
 - id

- name
- course
- age
- city

3. Insert at least **6 student records**.

Answer the all Questions

1. Display all student records.

```
//SELECT * FROM students;
```

2. Display only **student names and courses**.

```
//SELECT name,course FROM students;
```

3. Display students who live in **Colombo**.

```
//SELECT * FROM students WHERE city = 'Colombo';
```

4. Display students whose **age is greater than 21**.

```
//SELECT * FROM students WHERE age > 21;
```

5. Display students who follow the **IT course**.

```
//SELECT * FROM students WHERE course = 'IT';
```

Activity 2 - Library Book System

A small library wants to manage its book collection.

Each book has **BookID, Title, Author, Category, Price, and Quantity**.

Students must create the database and table and insert some records.

Tasks

1. Create a database called **LibraryDB**.

2. Create a table called **books** with fields:

- book_id
- title
- author

- category
- price
- quantity

3. Insert at least **6 books**.

Answer the all Questions

1. Display all books in the library.

```
//SELECT * FROM books;
```

2. Display only **title and price** of books.

```
//SELECT title,price FROM books;
```

3. Find books where the **price is greater than 1000**.

```
//SELECT * FROM books WHERE price >1000;
```

4. Find books that belong to the **Programming category**

```
//SELECT * FROM books WHERE category = 'Programing';
```

5. Display books where the **quantity is less than 10**.

```
//SELECT * FROM books WHERE quantity < 10;
```

Activity 3 - Employee Management System

A company wants to maintain employee records.

Each employee has **EmployeeID, Name, Department, Salary, Age, and City**.

Students must create the database and insert some sample data.

Tasks

1. Create a database named **CompanyDB**.

2. Create a table called **employees** with fields:

- employee_id
- name
- department
- salary
- age

- city

3. Insert at least **7 employees**.

Answer the all Questions

1. Display all employee details.

```
//SELECT * FROM employees
```

2. Display employee **name and salary** only.

```
//SELECT name,salary FROM employees;
```

3. Display employees who work in the **IT department**.

```
//SELECT * FROM employees WHERE department = 'IT';
```

4. Display employees whose **salary is greater than 50000**.

```
//SELECT * FROM employees WHERE salary > 50000;
```

5. Display employees sorted by **salary (ascending order)**.

```
//SELECT * FROM employees ORDER BY salary ASC;
```

6. Display employees sorted by **age (descending order)**.

```
//SELECT * FROM employees ORDER BY salary DESC;
```

Activity 4 - Product Sales System

A shop wants to manage product sales information.

Each product has **ProductID, ProductName, Category, Price, and Quantity**.

Students should calculate stock values using SELECT queries.

Tasks

1. Create a database called **SalesDB**.

2. Create a table named **products** with fields:

- product_id
- product_name
- category
- price

- quantity

3. Insert at least **6 products**.

Answer the all Questions

1. Display all products.\

```
//SELECT * FROM Products;
```

2. Display product **name and price**.

```
//SELECT product_name,price FROM products;
```

3. Display products with **price greater than 500**.

```
//SELECT * FROM products WHERE price > 500;
```

4. Display the **total value of each product**
(*price × quantity*).

```
//SELECT product_name,price,quantity,(price * quantity) AS total_value FROM products;
```

5. Display products sorted by **price (highest to lowest)**.

```
//SELECT * FROM products ORDER BY price DESC;
```

6. Display the **first 3 products only**.

```
//SELECT * FROM products LIMIT 3;
```

Activity 5 - Online Store Orders

An online store wants to track customer orders.

Each order contains **OrderID, CustomerName, ProductName, Price, Quantity, and OrderDate**.

Students must analyze order data using SELECT queries.

Tasks

1. Create a database called **OrderDB**.

2. Create a table named **orders** with fields:

- order_id
- customer_name
- product_name
- price

- quantity
- order_date

3. Insert at least **8 order records**.

Answer the all Questions

1. Display all orders.

```
//SELECT * FROM orders;
```

2. Display **customer name and product name** only.

```
//SELECT customer_name,product_name FROM orders;
```

3. Display orders where **quantity is greater than 2**.

```
//SELECT * FROM orders WHERE quantity > 2;
```

4. Display orders where **price is greater than 1000**.

```
//SELECT * FROM orders WHERE price > 1000;
```

5. Display the **total price of each order**
(*price × quantity*).

```
//SELECT customer_name,product_name,price,quantity,(price * quantity) AS total_price FROM orders;
```

6. Display orders sorted by **order date**.

```
//SELECT * FROM orders ORDER BY order_date ASC;
```

7. Display the **top 5 most recent orders**.

```
//SELECT * FROM orders ORDER BY order_date DESC LIMIT 5;
```

